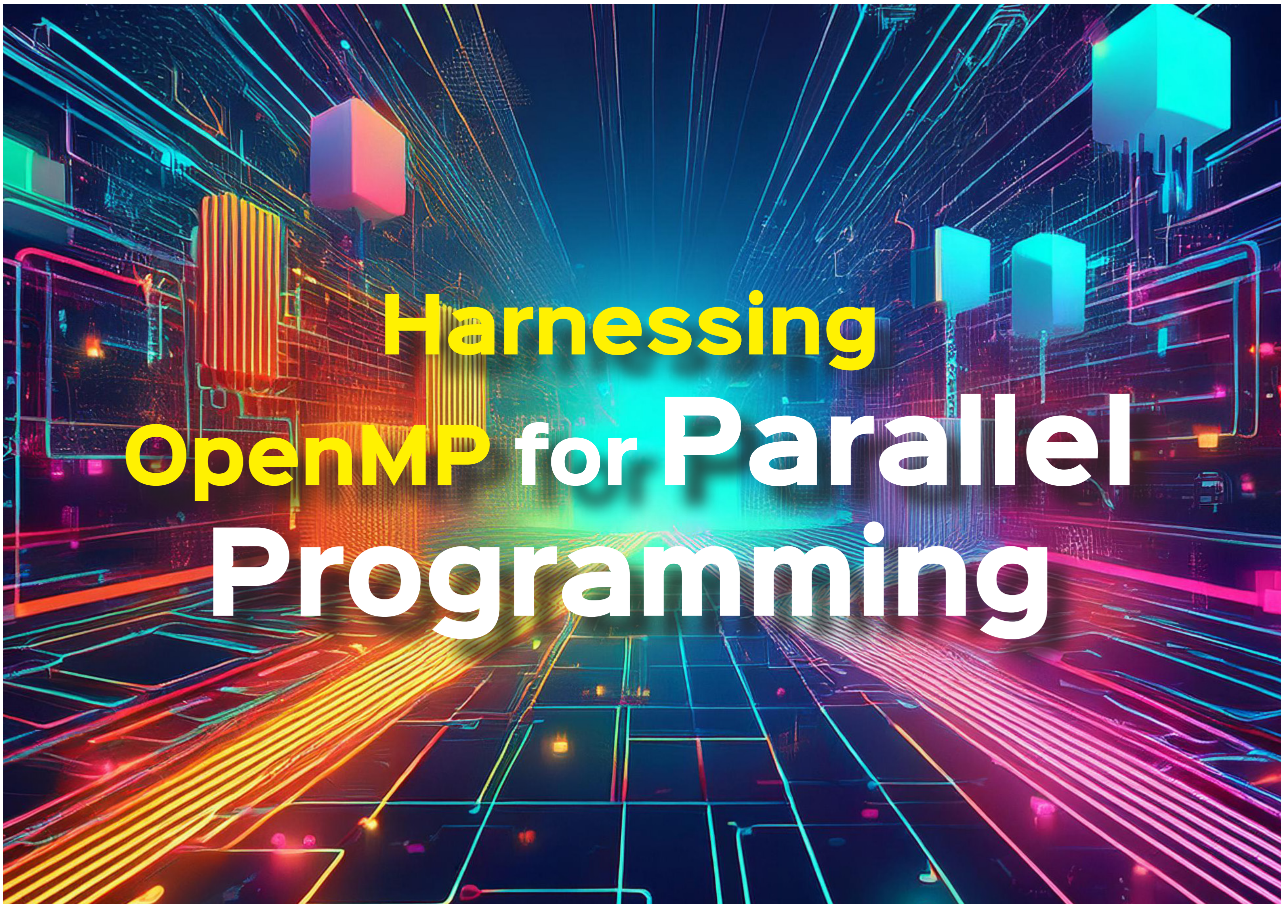


Harnessing OpenMP for Parallel Programming

Parallel programming has turned into a necessity with the vast amounts of data that organisations juggle with today. OpenMP is an open source application programming interface that offers a range of advanced features to make the task easier.

The processing needs of organisations have increased exponentially over the last few years, leading to the development of parallel computing to the next level. OpenMP is an open source application programming interface that helps in parallel programming. With the amount of data every business enterprise is receiving and needs to process, parallel computing is more like a necessity. OpenMP can help with that as it supports many platforms and multiple programming languages like C, C++ and Fortran. It provides a scalable and portable model that offers programmers more flexibility and easily understandable code.

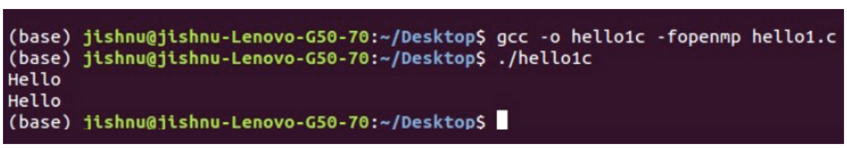
Before we understand some advanced features of OpenMP, let us look at a simple program to understand how it works.

You can paste the following code in a text editor:

```
#include<stdio.h>
#include<omp.h>
     int main()
{
     #pragma omp parallel
printf("Hello\n");
 }
```

Here we first import the headers required -- *omp.h* and *stdio.h.* Now, in the main code, write *#pragma omp* with the parallel keyword. This will run the code below it in parallel on different threads.

Next, save this text editor with a file name *.c.* Now open a terminal and type the following commands as shown in Figure 1.



```
(base) jishnu@jishnu-Lenovo-G50-70:~/Desktop$ gcc -o hello1c -fopenmp hello1.c
(base) jishnu@jishnu-Lenovo-G50-70:~/Desktop$ ./hello1c
Hello
Hello
(base) jishnu@jishnu-Lenovo-G50-70:~/Desktop$
```

Figure 1: Parallel processing